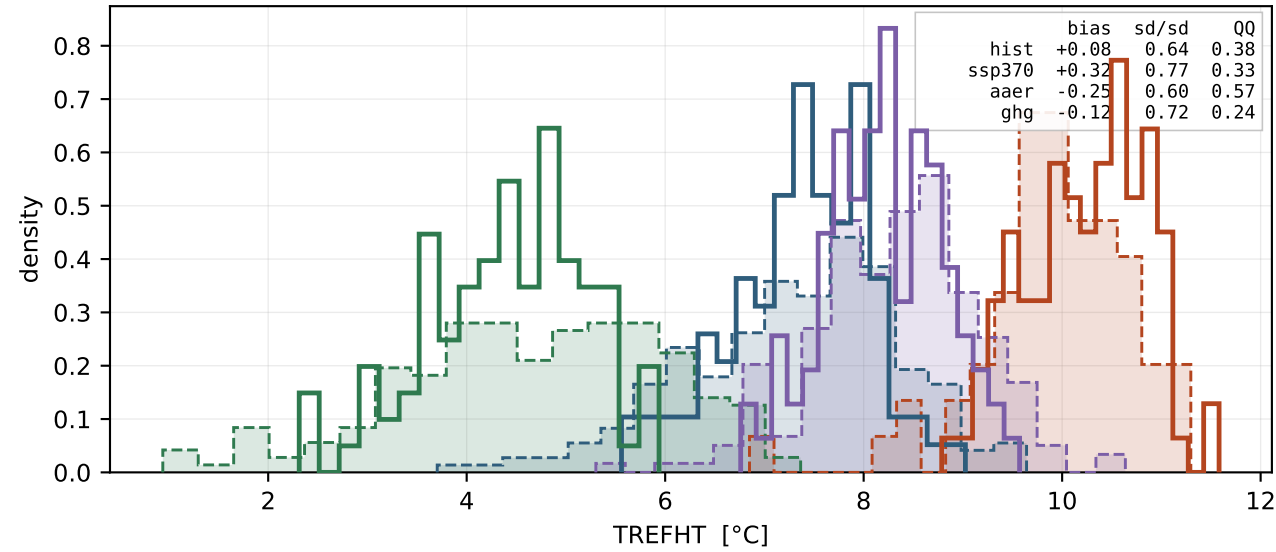
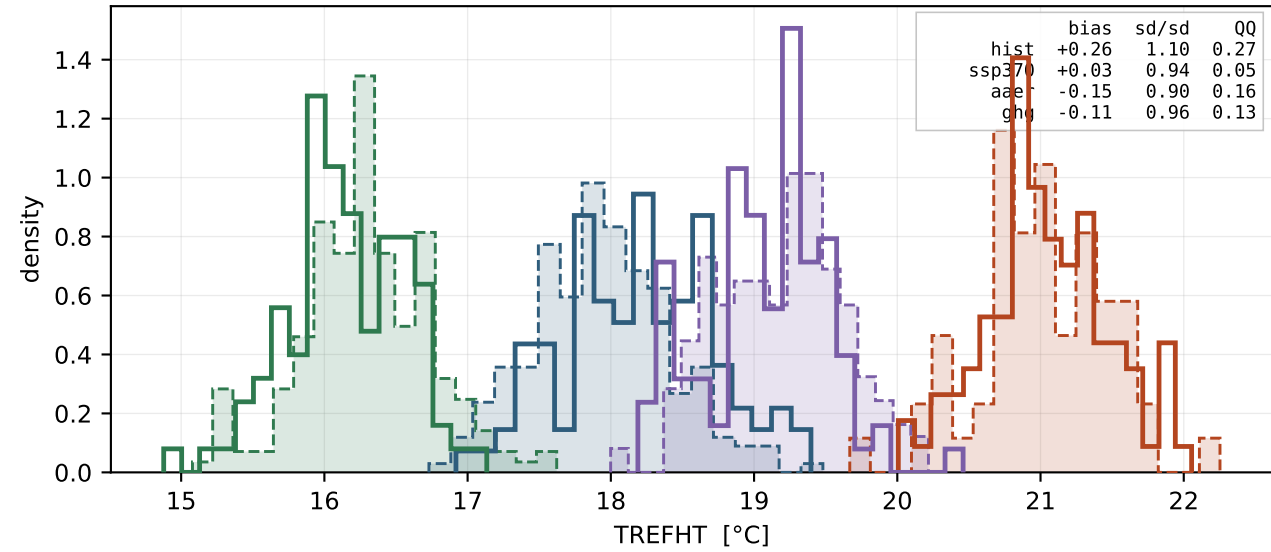


Distribution over the LAST 20 YEARS of each training experiment — step = emulator, filled/dashed = CESM2
 hist ends 2014, ssp370 and the single-forcing runs end 2100, so the panels compare different periods per experiment.
 All members pooled; the emulator contributes 5 x 20 samples, CESM2 up to 11 x 20.
 Inset: mean bias [°C], the sd ratio emulator/CESM2, and the Q-Q RMSE — the RMS gap between matched quantiles, which is a proper distance for unpaired samples where a pointwise RMSE would not be.

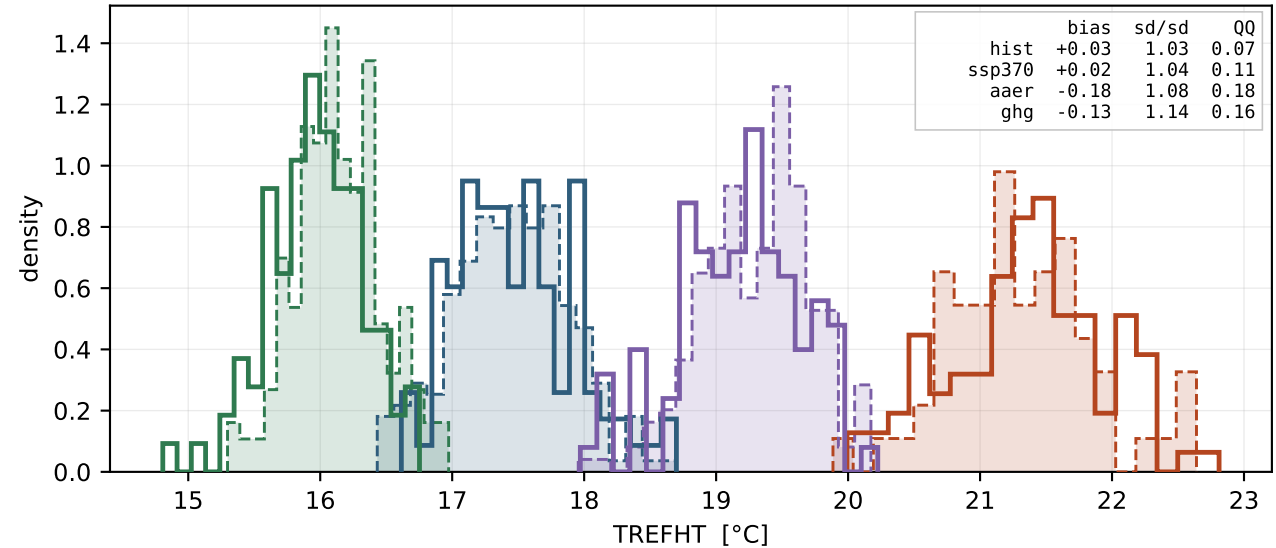
(a) Helsinki (cell +59.84, +25.00)



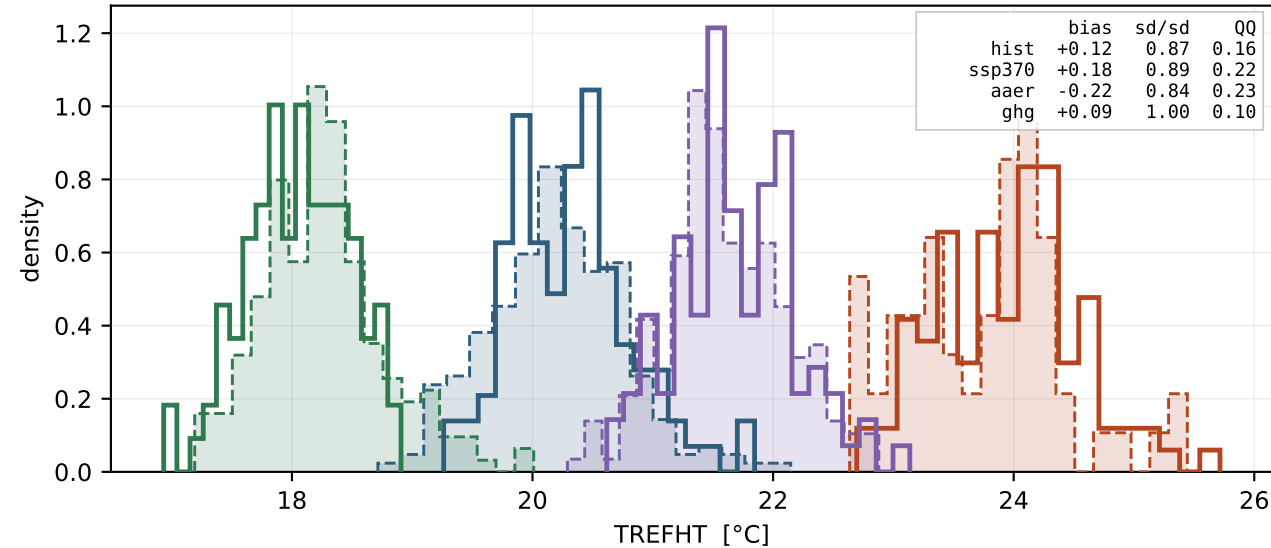
(b) Tokyo (cell +35.34, +140.00)



(c) Sydney (cell -33.46, +151.25)



(d) Sao Paulo (cell -23.09, -46.25)



— hist — emulator — ssp370 — emulator — aaer (GHG fixed at 1850) — emulator — ghg (aerosol fixed at 1850) — emulator
- - - hist — CESM2 - - - ssp370 — CESM2 - - - aaer (GHG fixed at 1850) — CESM2 - - - ghg (aerosol fixed at 1850) — CESM2